

Ridgeline HD Suspension Service Manual

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1 Introduction and Safety

This manual covers diagnosis and service of the Ridgeline HD suspension system, including struts, shocks, control arms, sway bar links, and wheel hub bearings. Always support the vehicle on rated stands and never work under a vehicle held only by a jack. Spring-loaded components store significant energy and require an approved spring compressor. For related undercarriage emissions routing, consult the Ridgeline HD Emissions Service Manual .

2 Suspension System Overview

The Ridgeline HD suspension comprises front strut assembly STRT-8500, rear shock absorber SHK-8520, control arm CARM-8540, sway bar link SBL-8560, and wheel hub bearing assembly WHB-8580. Together these components manage ride control, wheel location, and load handling for the heavy-duty platform. This architecture parallels the configuration documented in the Heron Suspension Service Manual . Inspect all components for wear before isolating a noise or handling complaint.

2.1 Front Strut Assembly

Front strut assembly STRT-8500 combines the damper and coil spring into a serviceable unit and locates the front wheel through the upper mount. Inspect STRT-8500 for fluid leakage, mount bearing binding, and worn isolators. A leaking or noisy STRT-8500 should be replaced as a complete assembly. Related coverage is in the Heron Suspension Service Manual :: Front Strut Sub-System .

2.2 Rear Shock Absorber

Rear shock absorber SHK-8520 dampens rear axle motion and is replaced in axle pairs to maintain balanced ride control. Inspect SHK-8520 for external fluid loss, bent rods, and worn bushings. Replace any SHK-8520 showing fade during a bounce test. See the Heron Suspension Service Manual :: Rear Shock Sub-System .

2.3 Control Arms and Sway Bar Links

Control arm CARM-8540 locates the wheel and carries the ball joint, while sway bar link SBL-8560 connects the stabilizer bar to the suspension. Inspect CARM-8540 bushings and the ball joint for play, and check SBL-8560 for worn end-link sockets. Worn SBL-8560 links commonly produce a rattle over rough pavement. Related guidance appears in the Heron Suspension Service Manual :: Control Arm and Linkage Sub-System .

2.4 Wheel Hub Bearing Assembly

Wheel hub bearing assembly WHB-8580 supports the wheel and integrates the speed sensor tone ring. Inspect WHB-8580 for roughness, axial play, and wheel speed sensor faults. Replace WHB-8580 as a unit; the bearing is not separately serviced. See the Heron Suspension Service Manual :: Wheel Hub Bearing Sub-System .

3 Front Strut Sub-System Service

Service of the front strut covers assembly STRT-8500 and addresses ride-height, noise, and damping complaints at the front axle. Always note alignment settings before disassembly, as strut service affects camber and caster. Use a spring compressor whenever separating the spring from the damper. Refer to the Heron Suspension Service Manual :: Front Strut Sub-System for shared procedures.

3.1 Front Strut Replacement Procedure

Follow PROC-STRUT to support the lower control arm, disconnect the sway bar link and any sensor brackets, and remove the strut-to-knuckle bolts. Lower the assembly and remove STRT-8500 from the upper tower nuts. Install the new STRT-8500, torque all fasteners to specification, and perform a four-wheel alignment. The matching procedure is the Heron Suspension Service Manual :: Front Strut Replacement .

4 Control Arm and Sway Bar Sub-System Service

This sub-system covers control arm CARM-8540 and sway bar link SBL-8560, which together influence wheel location and roll control. Inspect both components whenever a clunk or wander complaint is raised. Replace bushings and links in axle sets where wear is found. Related coverage is in the Heron Suspension Service Manual :: Control Arm and Linkage Sub-System .

4.1 Control Arm Replacement Procedure

Follow PROC-CONTROL-ARM to separate the ball joint from the knuckle, disconnect sway bar link SBL-8560, and remove the CARM-8540 pivot bolts. Install the new CARM-8540, leaving pivot fasteners loose until the suspension is loaded to ride height. Torque all fasteners at ride height and align the vehicle. See the Heron Suspension Service Manual :: Sway Bar Link Replacement for the companion link procedure.

5 Wheel Hub Bearing Sub-System Service

Service of the wheel hub bearing covers assembly WHB-8580 and addresses bearing noise, play, and associated wheel speed sensor faults. Always inspect the mating knuckle bore for corrosion before installing a new assembly. Confirm the tone ring is clean and undamaged. Refer to the Heron Suspension Service Manual :: Wheel Hub Bearing Sub-System .

5.1 Wheel Hub Bearing Replacement Procedure

Follow PROC-HUB-BEARING to remove the brake caliper, rotor, and axle nut, then unbolt WHB-8580 from the knuckle. Clean the bore, install the new WHB-8580, and torque the mounting bolts and axle nut to specification. Verify wheel speed sensor operation before returning the vehicle to service. The parallel procedure is the Heron Suspension Service Manual :: Wheel Hub Bearing Replacement .

6 Rear Shock Sub-System Service

Service of the rear shocks covers absorber SHK-8520 and addresses harsh ride, bottoming, and rear-end float complaints. Replace SHK-8520 in axle pairs and torque the upper and lower mounts at ride height. Inspect the mounting bushings and brackets for elongation while the shock is removed. See the Heron Suspension Service Manual :: Rear Shock Absorber Replacement .

7 Torque Specifications and Reference

Torque all strut, shock, control arm, sway bar link, and wheel hub bearing fasteners to the values listed in the specification table using a calibrated wrench. Many suspension fasteners are single-use and must be replaced during service. Always perform a four-wheel alignment after any procedure affecting wheel location. For adjacent body fastener references, see the Ridgeline HD Body Service Manual .